THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE UNIT TEST SET 3

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The ovule develops into a 1

(a) petal **(b)** fruit **(c)** flower **(d)** seed

2. In the dihybrid F2, the number of round green seeds out of 16 is 1

(a) 3 **(b)** 4 **(c)** 9 **(d)** 1

3. The filtration of blood in the nephron takes place in: 1

(a) collecting duct **(b)** ureter **(c)** tubule **(d)** glomerulus

4. Dwarfism is caused by a deficiency of: 1

(a) growth hormone **(b)** adrenaline **(c)** thyroxine **(d)** insulin

5. The ova (eggs) are produced in 1

(a) oviduct **(b)** uterus **(c)** vagina **(d)** ovaries

6. A normal human male has the sex chromosomes 1

(a) YY **(b)** XY **(c)** XXY **(d)** XX

7. The breakdown of pyruvate to give carbon dioxide, water and energy takes place in: 1

(a) chloroplast **(b)** cytoplasm **(c)** nucleus **(d)** mitochondria

8. The hormone which helps in the growth of the shoot tip towards light is: 1

(a) gibberellin **(b)** auxin **(c)** abscisic acid **(d)** cytokinin

9. Rhizopus reproduces by 1

(a) fragmentation **(b)** binary fission **(c)** buds **(d)** spores

10. When pure tall (TT) and pure dwarf (tt) pea plants are crossed, the F1 generation is 1

(a) all dwarf **(b)** three tall to one dwarf
(c) all tall **(d)** half tall and half dwarf

11. The energy currency of the cell is: 1

(a) starch **(b)** ATP **(c)** glucose **(d)** ADP

12. Growth of the pollen tube towards the ovule is an example of: 1

(a) phototropism **(b)** geotropism **(c)** thigmotropism **(d)** chemotropism

13. Transfer of pollen from anther to stigma is called 1

(a) fertilisation **(b)** pollination **(c)** germination **(d)** fission

14. The new combinations of traits (round green and wrinkled yellow) in F2 show that 1

(a) no dominant trait exists **(b)** F1 gets only one parent's traits

(c) traits are inherited independently **(d)** traits are always linked

15. Artificial kidney works on the principle of: 1

(a) dialysis **(b)** filtration only **(c)** osmosis only **(d)** photosynthesis

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. What is the function of the placenta? 2

17. What is an acquired trait? Give an example. 2

18. Where do plants get each of the raw materials required for photosynthesis? 2

19. What happens at the synapse between two neurons? 2

20. How does binary fission differ from multiple fission? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Why is DNA copying essential in reproduction? 3

22. How do Mendel's experiments show that traits are inherited independently? 3

23. Why are the alveoli covered with a network of blood vessels? 3

24. How does the sensitive plant respond to touch? 3

OR

Explain the changes that occur in a flower after fertilisation. 3

25. Differentiate between dominant and recessive traits with the help of a pea example. 3

OR

Why do we feel cramps in muscles after vigorous exercise? Name the substance. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the human alimentary canal with the role of each part. 5

OR

Explain reflex arc with a labelled description and give three differences between reflex action and voluntary action. 5

27. Describe binary fission in Amoeba. Why is Leishmania's division different? 5

OR

Describe Mendel's dihybrid cross using round yellow and wrinkled green seeds. Give the phenotypes and ratio in F2. 5

END OF QUESTION PAPER — Total: 50 Marks